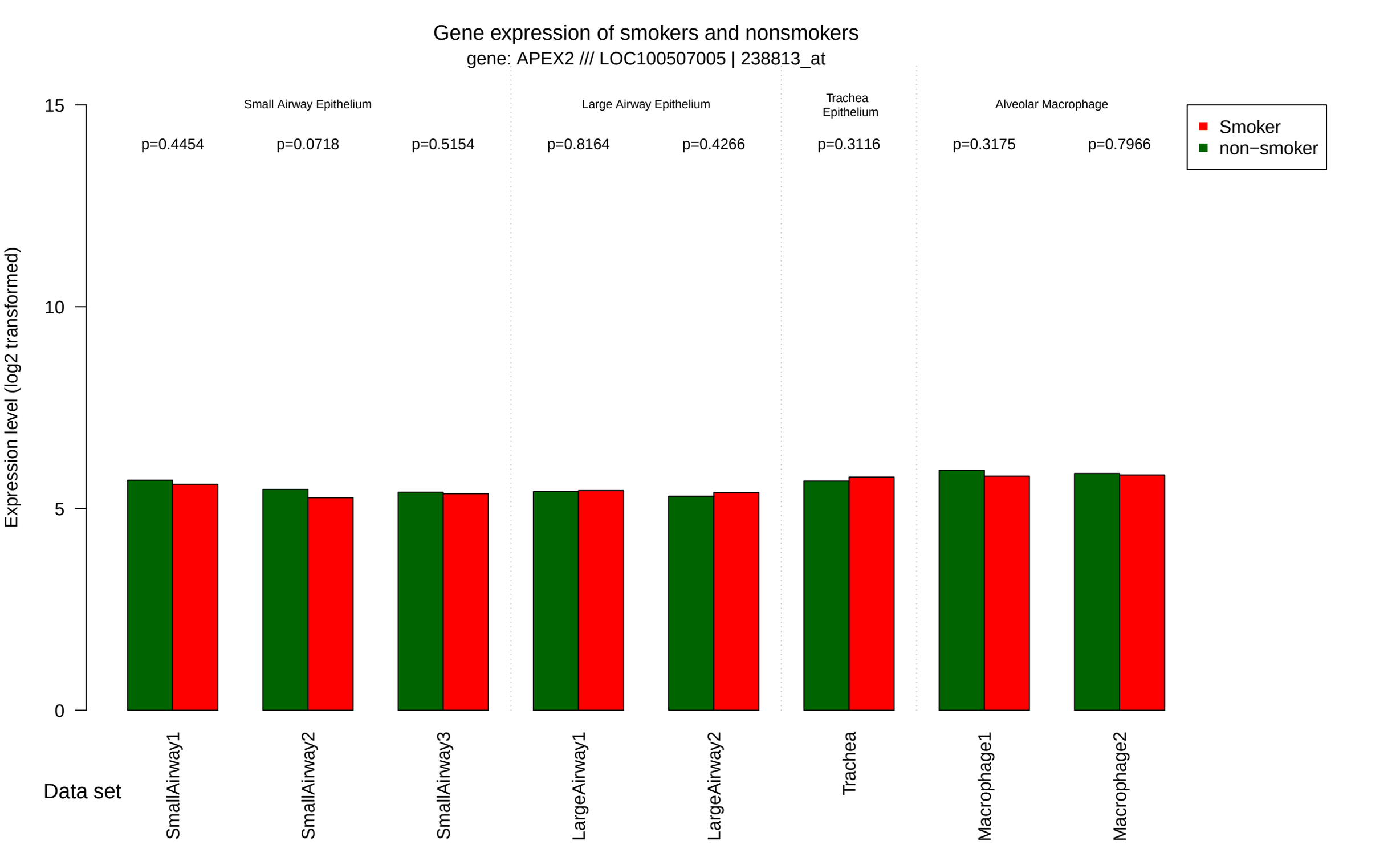
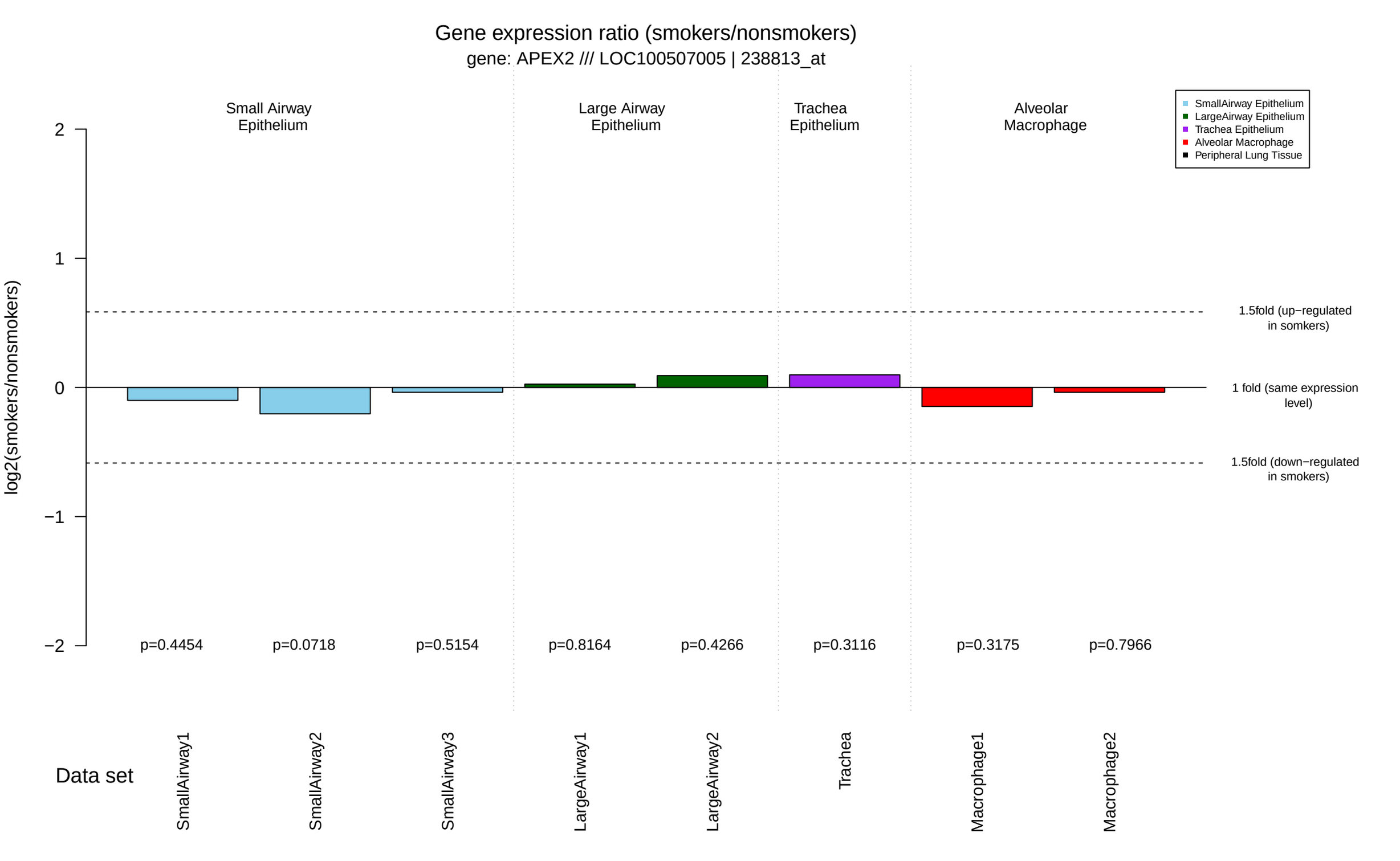
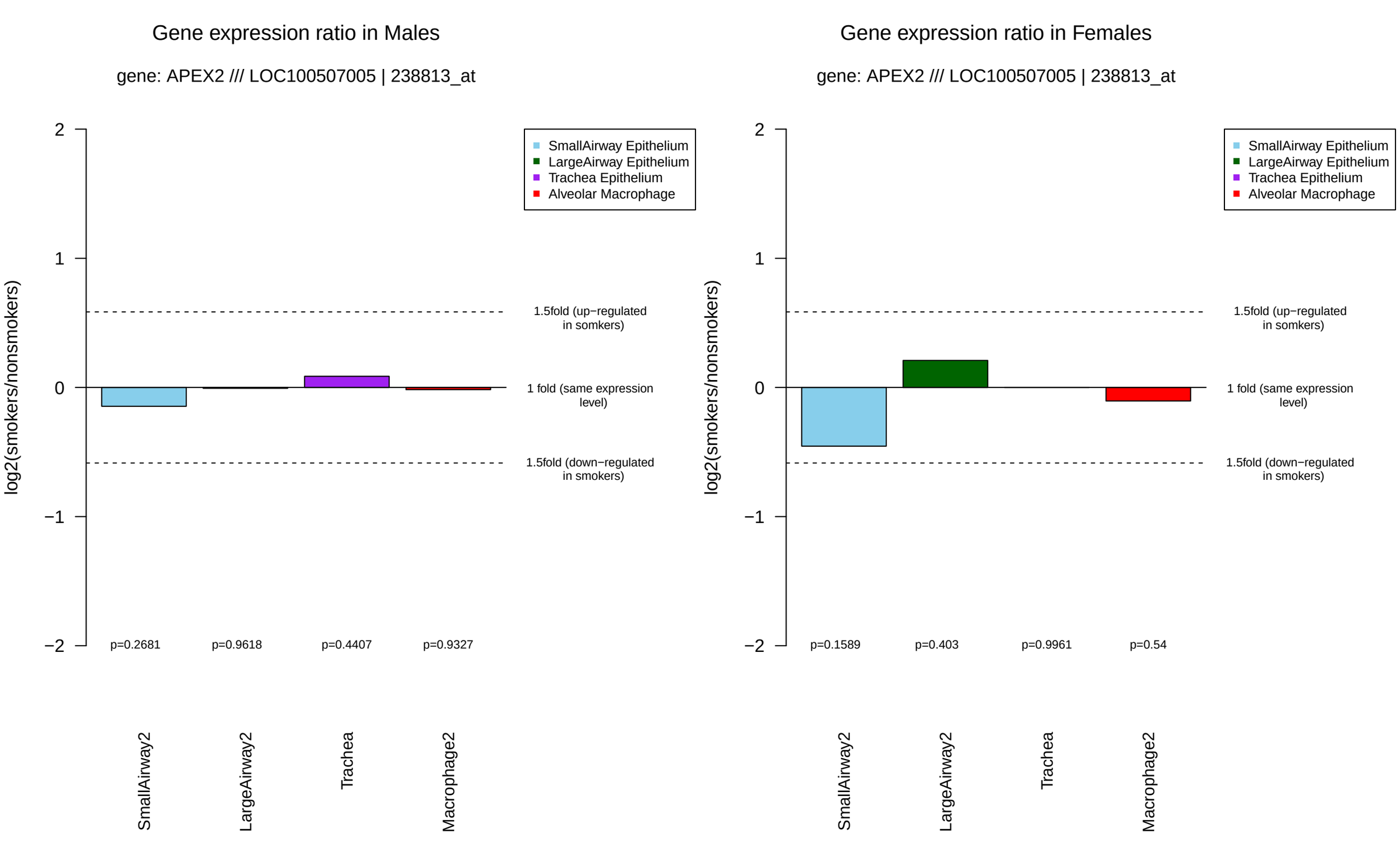
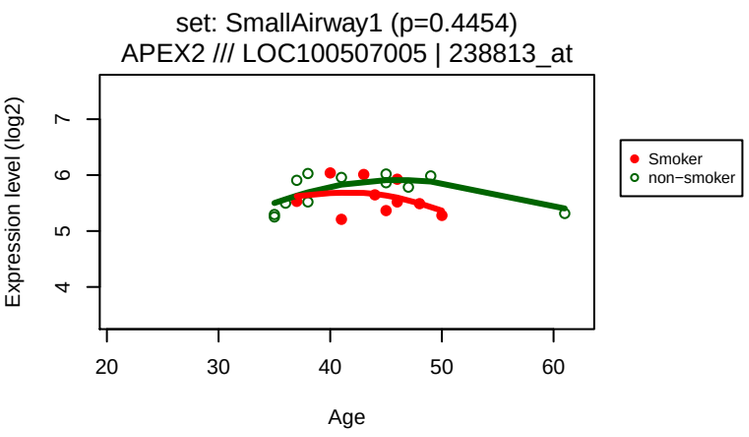
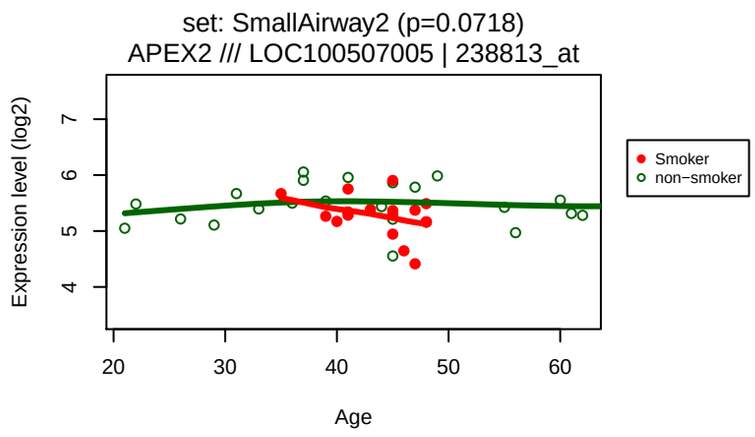
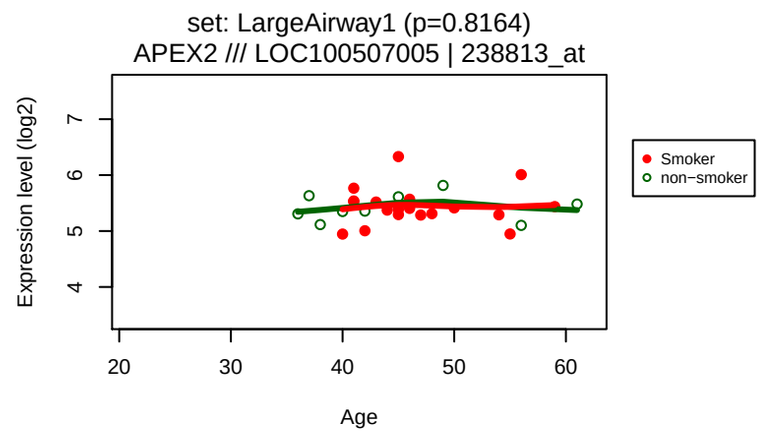

Gene expression of smokers and nonsmokers
gene: APEX2 /// LOC100507005 | 238813_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4454
p=0.0718
p=0.5154
p=0.8164
p=0.4266
p=0.3116
p=0.3175
p=0.7966
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

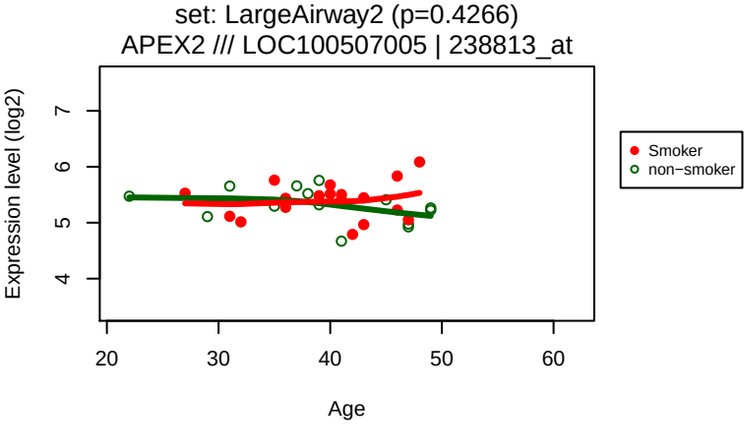

Gene expression ratio (smokers/nonsmokers)
gene: APEX2 /// LOC100507005 | 238813_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4454
p=0.0718
p=0.5154
p=0.8164
p=0.4266
p=0.3116
p=0.3175
p=0.7966
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

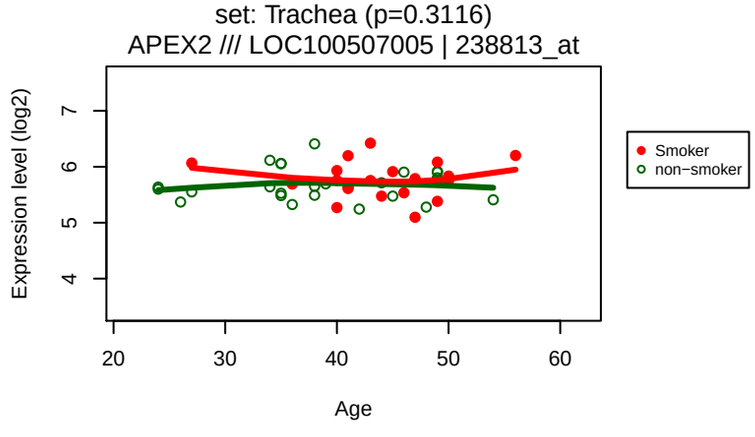

Gene expression ratio in Males
gene: APEX2 /// LOC100507005 | 238813_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2681
p=0.9618
p=0.4407
p=0.9327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APEX2 /// LOC100507005 | 238813_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1589
p=0.403
p=0.9961
p=0.54
SmallAirway2
LargeAirway2
Trachea
Macrophage2

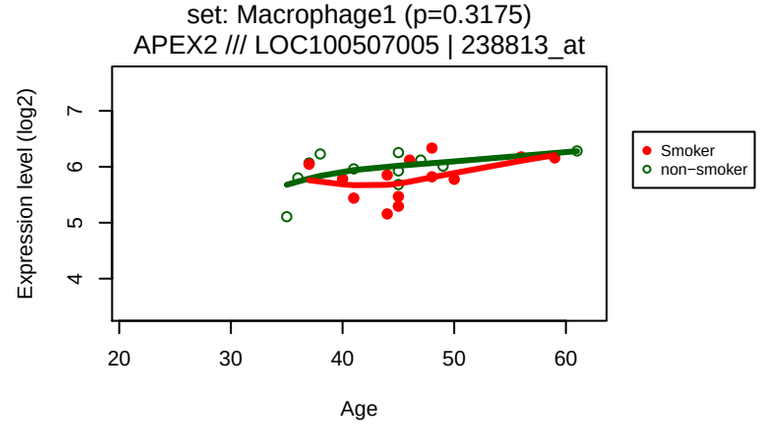

set: SmallAirway1 (p=0.4454)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker

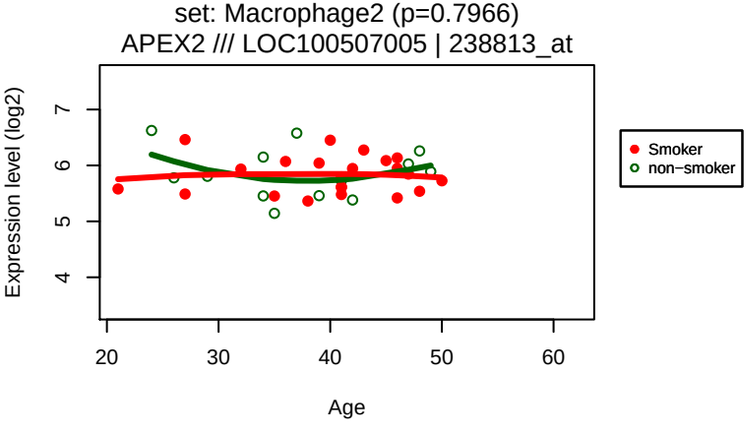

set: SmallAirway2 (p=0.0718)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker

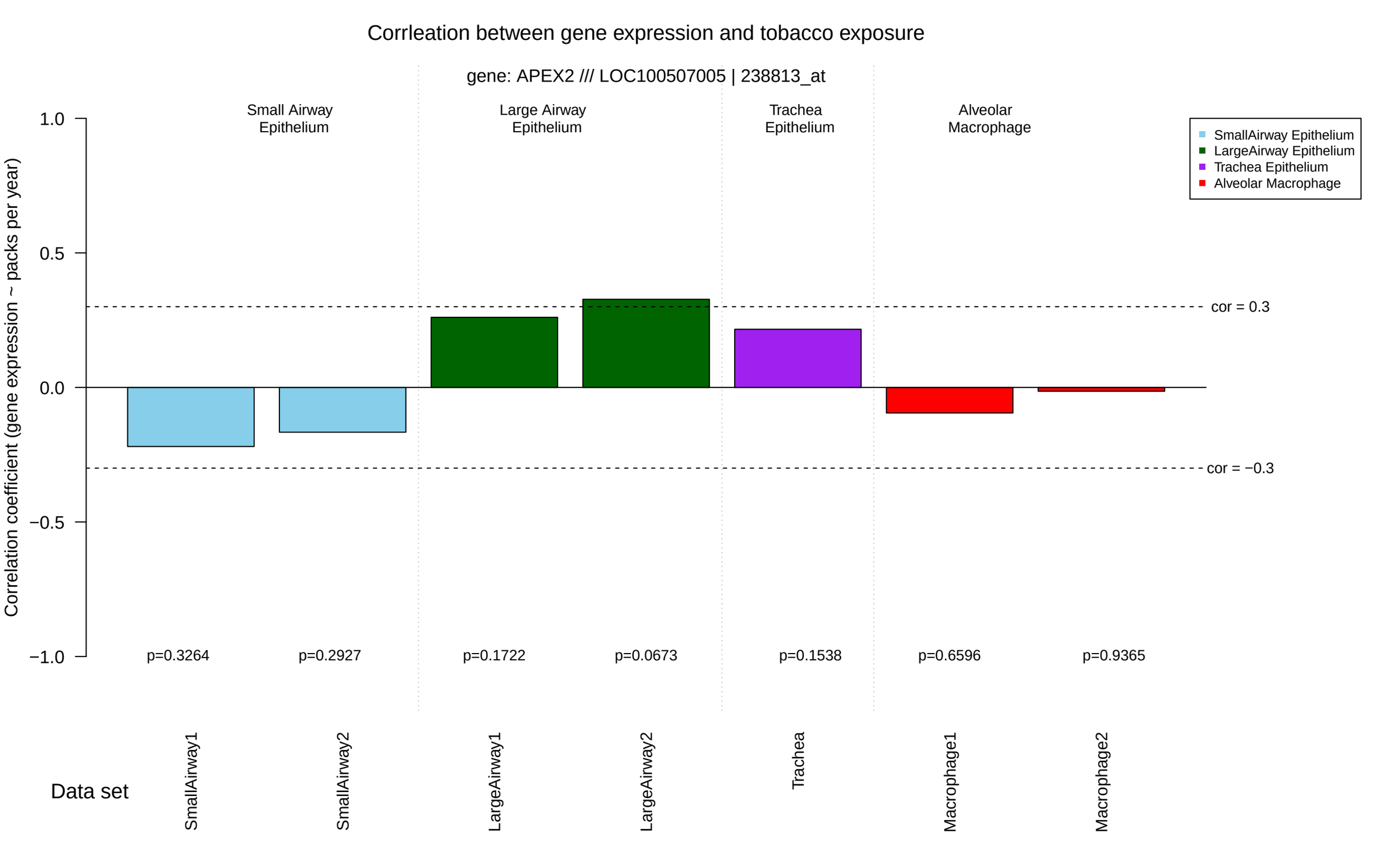

set: LargeAirway1 (p=0.8164)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4266)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3116)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3175)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7966)
APEX2 /// LOC100507005 | 238813_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APEX2 /// LOC100507005 | 238813_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3264
p=0.2927
p=0.1722
p=0.0673
p=0.1538
p=0.6596
p=0.9365
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2